Hw.3

A) I $P(A_1) = P(A_2) = P(A_3) = 1/4$

$P((A_1^c \cap A_2^c) \cup A_3) = P(A_1^c \cap A_2^c) + P(A_3) - P((A_1^c \cap A_2^c) \cap A_3)$

$P(A_1^c) = P(A_2^c) = 1 - 1/4 = \frac{3}{4}$

$P(A_1^c \cap A_2^c) = P(A_1^c) \times P(A_2^c)$ {∵ Independent events complement are also independent}

$= \frac{3}{4} \times \frac{3}{4}$

$= 9/16$

$P((A_1^c \cap A_2^c) \cap A_3) = P(A_1^c \cap A_2^c \cap A_3)$

∵ mutually Independent we can directly multiply

$= P(A_1^c) \times P(A_2^c) \times P(A_3)$

$= \frac{3}{4} \times \frac{3}{4} \times \frac{1}{4} = \frac{9}{64}$

$P((A_1^c \cap A_2^c) \cup A_3) = \frac{9 \times 4}{16 \times 4} + \frac{1 \times 16}{4 \times 16} - \frac{9}{64} = \frac{36+16-9}{64}$

$\boxed{= \frac{43}{64}}$

A2) To prove if $P(B|A) = P(B|A^c)$ then A & B are independent

if independent $P(A \cap B) = P(A) . P(B)$

and
$$P(B|A) = \frac{P(B \cap A)}{P(A)}$$ conditional probability

Law of total probability

$$P(B) = P(B|A) . P(A) + P(B|A^c) . P(A^c)$$

$$= P(B|A) . P(A) + P(B|A) . P(A^c)$$

since $P(B|A) = P(B|A^c)$

$$= P(B|A) (P(A) + P(A^c))$$

$$P(B) = P(B|A) \quad (I)$$

$$P(B) = \frac{P(A \cap B)}{P(A)}$$

$$\therefore \quad P(A \cap B) = P(A) . P(B)$$

thus independent events.

Intuitive

The probability of event B happening is exact same whether event A happens

Or event A does not happen.

This means that knowing event A outcomes Provides absolutely no new information about the Likelyhood of event B.

If information about A doesn't change the P(B), then A & B must be independent.

A3 (a)

$$P(Bowl_1) = P(Bowl_2) = 1/2$$

$$P(Red \mid Bowl_1) = 3/10$$
$$P(Red \mid Bowl_2) = 6/10$$

$$P(Red) = P(Red \mid Bowl_1) \cdot P(Bowl_1) + P(Red \mid Bowl_2) \cdot P(Bowl_2)$$

$$= \left(\frac{3}{10} \times \frac{1}{2}\right) + \left(\frac{6}{10} \times \frac{1}{2}\right)$$

$$= \left(\frac{9}{20}\right)$$

b) $P(Bowl_2 \mid Red)$

Using Baye's

$$P(Bowl_2 \mid Red) = \frac{P(Red \mid Bowl_2) \cdot P(Bowl_2)}{P(Red)}$$

$$= \frac{\frac{6}{10} \times \frac{1}{2}}{\frac{9}{20}} = \frac{\frac{6}{\cancel{20}}}{\frac{9}{\cancel{20}}} = \frac{2}{3}$$

c) for independent

$$P(A \cap B) = P(A) \cdot P(B)$$

let's calculate both

$$P(\text{Red} \cap \text{Bowl}_2) = P(\text{Red} \mid \text{Bowl}_2) \cdot P(\text{Bowl}_2)$$

$$\text{LHS} = \frac{6}{10} \times \frac{1}{2} = \frac{6}{20} = \frac{3}{10}$$

$$\text{RHS} = P(\text{Red}) \cdot P(\text{Bowl}_2)$$

$$= \frac{9}{20} \times \frac{1}{20} = \frac{9}{40}$$

since $\frac{3}{10} \neq \frac{9}{40}$ $\therefore$ not independent

A4) $P(x) = c\left(\frac{2}{3}\right)^x \quad , x = 1, 2, \ldots$

2 conditions

1) non negative probability
2) they should all add up to 1

$P(1) = \frac{2}{3}c \quad P(2) = c\left(\frac{2}{3}\right)^2 \ldots \quad c\sum_{x=1}^{\infty}\left(\frac{2}{3}\right)^x = 1$

$$c\left(\frac{2}{3} + \left(\frac{2}{3}\right)^2 + \left(\frac{2}{3}\right)^3 + \ldots\ldots\right)$$

Let $2/3 = r$

$$c\left(r + r^2 + r^3 + \ldots\right)$$

$$c\left(\frac{r}{1-r}\right) = c\left(\frac{2/3}{1-2/3}\right) = 1$$

$$\frac{2/\cancel{3}}{1/\cancel{3}} \quad 2c = 1 \quad \boxed{c = 1/2 = 0.5}$$

b) $P(x) = cx \quad , x = 1, 2, 3, 4, 5, 6$

$$\underbrace{c + 2c + 3c + 4c + 5c + 6c}_{\frac{n(n+1)}{2}} = 1$$

$$c\,(1+2+3+\ldots 6) = 1$$

$$c \cdot \frac{\overset{3}{\cancel{6}} \times \overset{7}{\cancel{14}}}{\cancel{2}} = 1 \quad \boxed{c = \frac{1}{21}}$$

$$\boxed{c = 1/21}$$

A5 (a) $P(X \leq 50)$

$$= \sum_{x=1}^{50} P(x) = \sum_{x=1}^{50} \frac{x}{5050}$$

$$\frac{1}{5050} \sum_{x=1}^{50} x = \frac{1}{5050} \times \frac{50 \times 51}{2} = \frac{1275}{5050}$$

$$= \frac{51}{202}$$

b)

$$F(x) = P(X \leq x) = \sum_{k=1}^{[x]} \frac{k}{5050}$$

$$= \frac{1}{5050} \sum_{k=1}^{[x]} k = \frac{1}{}$$

first n int sum

$$= \frac{1}{5050} \left(\frac{[x][x]+1)}{2} \right) = \frac{[x]([x]+1)}{10100}$$

~~F(x)~~ for x in $[1, 100]$

$$F(1) = \frac{(1)(1+1)}{10100} = \frac{2}{10100} = \frac{1}{5050}$$

$$F(100) = \frac{(100)(101)}{10100} = 1$$

for absdute upperlimit of range $x = 100$
CDF should account for all possible outcomes $(X \leq 100)$ so it should be 1

~~Beyond max value $(X > 100)$~~

~~Let's try $x = 150$~~

~~$F(150) = P(X \leq 150)$~~

A6) $P(x) = \frac{1}{2^x}$, $x = 1, 2 \cdots$ PMF of X

Let $Y = X^3$

a) $P(Y = y) = P(g(X) = y)$

Since X can be $\{1, 2, 3, 4, \dots\}$
support of $Y = \{1^3, 2^3, 3^3, 4^3, \dots\}$
$= \{1, 8, 27, 64, \dots\}$

Relating Y back to x

$y = x^3 \qquad x = \sqrt[3]{y}$

$P_Y(y) = P(Y = y) = P(X = \sqrt[3]{y}) = P_X(\sqrt[3]{y})$

Subsituting into PMF of X

$$P_Y(y) = \left(\frac{1}{2}\right)^{\sqrt[3]{y}}$$

The PMF of Y $p_Y(y) = \left(\frac{1}{2}\right)^{\sqrt[3]{y}}$

for $y \in \{1, 8, 27, 64, \dots\}$ = 0 otherwise

(b) finding CDF of Y

$$F_Y(y) = P(Y \leq y)$$

$$= P(X^3 \leq y)$$

$$= P(X \leq \sqrt[3]{y})$$

finding the CDF of X, $F_X(k) = P(X \leq k)$

we need to sum PMF of X from $i=1$ upto k

$$F_X(k) = \sum_{i=1}^{k} \left(\frac{1}{2}\right)^{i}$$

this is finite GP series formula $\sum_{i=1}^{k} r^i = r \cdot \frac{1-r^k}{1-r}$

$$= \frac{1}{2} \cdot \frac{1-(1/2)^k}{1-1/2} = \frac{1}{2} \cdot \frac{1-(1/2)^k}{1/2} =$$

$$= 1 - \left(\frac{1}{2}\right)^k$$

Combining the result.

$$F_Y(y) = P(X \le \sqrt[3]{y})$$

since X only takes integer values, Probability only increases at integer. so floor function

$$F_Y(y) = F_X(\lfloor \sqrt[3]{y} \rfloor) = 1 - \left(\frac{1}{2}\right)^{\lfloor \sqrt[3]{y} \rfloor}$$

CDF of Y is $F_Y(y) = 1 - \left(\frac{1}{2}\right)^{\lfloor \sqrt[3]{y} \rfloor}$ for $y \ge 1$

& $F_Y(y) = 0$ for $y < 1$

A7) Total outcomes

Choosing 2 distinct integers from 6 $\binom{6}{2} = 15$

X = absolute possible differences $\{1, 2, 3, 4, 5\}$

$X = 1$ $\{1,2\}, \{2,3\}, \{3,4\}, \{4,5\}, \{5,6\}$ 5 Pairs
$X = 2$ $\{1,3\}, \{2,4\}, \{3,5\}, \{4,6\}$ 4 pairs.
$X = 3$ $\{1,4\}, \{2,5\}, \{3,6\} \to$ 3 Pairs.
$X = 4$ $\{1,5\}, \{2,6\} \to$ 2 Pairs
$X = 5$ $\{1,6\}$ 1 Pairs

PMF, $P(X)$
$P(1) = 5/15$
$P(2) = 4/15$
$P(3) = 3/15$
$P(4) = 2/15$
$P(5) = 1/15$

$$E(x) = 1 \cdot \frac{5}{15} + 2 \cdot \frac{4}{15} + 3 \cdot \frac{3}{15} + 4 \cdot \frac{2}{15} + 5 \cdot \frac{1}{15}$$

$$\boxed{= 7/3}$$

Var(x)

$$E(x^2) = \sum x^2 \cdot P(x)$$

$$1^2 \cdot \frac{5}{15} + 2^2 \cdot \frac{4}{15} + 3^2 \cdot \frac{3}{15} + 4^2 \cdot \frac{2}{15} + 5^2 \cdot \frac{1}{15}$$

$$E(x^2) = \frac{105}{15} = 7$$

$$Var(X) = E(x^2) - (E(x))^2$$

$$= 7 - \left(\frac{7}{3}\right)^2 = \frac{14}{9}$$

A8) Linearity of expectation

$$E(aX + bY + c) = aE(X) + bE(Y) + c$$

we apply this rule to $U = 2X - Y + 1$

$E(U) = E(2X - Y + 1) = 2E(X) - E(Y) + E(1)$

$E(U) = 2(5) - 3 + 1$

$\boxed{= 8}$

Var(U)

2 key Properties of independent random variables

1) $Var(Z + c) = Var(Z)$

↳ adding constant doesn't change Varianc

2) for Linear combination coefficient are squared

$Var(aX \pm bY) = a^2 Var(X) + b^2 Var(Y)$

we apply this rule to

$Var(U) = Var(2X - Y + 1) = Var(2X - Y)$

$= (2)^2 Var(X) + (-1)^2 Var(Y)$

$= 4 \times (9) + 1(25)$

$Var(U) = 36 + 25 = 61$

A9) we want to show

$$E(X) = \sum_{K=0}^{\infty} [1 - F(K)]$$

RHS $F(K)$ is CDF where $F(K) = P(X \leq K)$

$\therefore$ $1 - F(K)$ is probability that X is greater than K

$$RHS = \sum_{K=0}^{\infty} P(X > K)$$

2) Express $P(X > K)$ as a sum of probabilities from PMF

event $X > K$ is same as event that X is $K+1$, ~~or~~ $K+2$, or $K+3$

$$P(X > K) = P(K+1) + P(K+2) + \ldots \ldots = \sum_{j=K+1}^{\infty} P(j)$$

substitute back into main sum

$$RHS = \sum_{K=0}^{\infty} \left(\sum_{j=K+1}^{\infty} P(j) \right)$$

Lets write first few terms

$k=0 \quad P(1) + P(2) + P(3) + P(4) + \cdots$

$k=1 \quad P(2) + P(3) + P(4) + \cdots$

$k=2 \quad P(3) + P(4) + \cdots$

$k=3 \quad P(4) + \cdots$

and so on

Instead of summing each row first, we can sum column

$P(1)$	once
$P(2)$	twice
$P(3)$	3 times
$\vdots$	$\vdots$

$$RHS = P(1) + 2 \cdot P(2) + 3P(3) + \cdots = \sum_{j=1}^{\infty} j \cdot P(j)$$

Standard def of expected value is

$$E(x) = \sum_{j=0}^{\infty} j \cdot P(j)$$

$$E(x) = (0 \cdot P(0)) + (1 \cdot P(1)) + (2 \cdot P(2)) + \cdots$$

Since I term is 0

$$\boxed{E(x) = \sum_{j=1}^{\infty} j \cdot P(j)}$$

$\therefore \quad E(X) = \sum_{K=0}^{\infty} [1 - F(K)]$

simplifies to $\sum_{j=1}^{\infty} j \cdot P(j)$

A1) $P(X) = \frac{1}{5} \qquad X = 1, 2, 3, 4, 5.$

a) $E(X) = 1 \times \frac{1}{5} + 2 \times \frac{1}{5} + 3 \times \frac{1}{5} + 4 \times \frac{1}{5} + 5 \times \frac{1}{5}$

$\therefore$ P(X) = is constant

$= \frac{15}{5} = \boxed{3}$

b) Var(X) =

$E(X^2) = \sum_{x=1}^{5} x^2 \cdot P(x)$

$\left(1^2 \cdot \frac{1}{5}\right) + \left(2^2 \cdot \frac{1}{5}\right) + \left(3^2 \cdot \frac{1}{5}\right) + \left(4^2 \cdot \frac{1}{5}\right)$

$+ \left(5^2 \cdot \frac{1}{5}\right)$

$= \frac{55}{5} = 11$

$Var(x) = E(x^2) - (E(x))^2$

$= 11 - 9$

$\boxed{= 2}$

(c) $E[(x+2)^2]$

$E[x^2 + 4x + 4]$

$= E[x^2] + E[4x] + E(4)$

$= E(x^2) + 4E(x) + 4$

$= 11 + 4 \cdot 3 + 4$

$= 11 + 12 + 4$

$= \boxed{27}$